

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (CL) Examination

November-December 2012

CL203 Fluid Mechanics

Date: 29.11.2012, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Define Path line, Steady flow and Rotational flow. [03]
- (b) A stream function is given by $\psi = 8x - 10y$. Determine the velocity components and also magnitude and direction of the resultant velocity at any point. [04]
- Q - 2 (a) Define Fluid. Differentiate between Liquid and Gas. [05]
- (b) Enlist limitations of piezometer. [02]
- (c) A circular plate of diameter 1 m is immersed in a liquid of specific gravity 0.9 with its plane making an angle of 30° with the horizontal. The centre of the plate is at a depth of 2m below the free surface. Calculate (i) total pressure force on one side of the plate and (ii) location of the centre of pressure. [03]
- (d) A block of width 1.5m, depth 1m and length 4 m is floating in water. If the density of block is 700 kg/m^3 , find the volume of water displaced and the position of centre of buoyancy. [04]

OR

- Q - 2 (a) State and prove Pascal's Law. [05]
- (b) Define Viscosity and Kinematic viscosity. [05]
- Find kinematic viscosity of an oil having density 980 kg/m^3 when at a certain point in the oil, the shear stress is 0.25 N/m^2 and velocity gradient 0.3 per second.
- (c) Explain Atmospheric pressure, Gauge pressure and Vacuum pressure. Also show the relation between them with neat sketch. [04]
- Q - 3 (a) Derive the continuity equation for three dimensional flow. [05]
- (b) A rectangular orifice 0.8m wide and 1m deep is discharging water from a vessel. The top edge of the orifice is 0.5m below the water surface in the vessel. Determine: (i) the discharge through the orifice if $C_d = 0.63$, and (ii) the percentage error if the orifice is treated as a small orifice. [05]

- (c) Give the detailed classification of mouthpiece. [04]

OR

- Q - 3 (a) Define Forced vortex flow and free vortex flow. Discuss the free vortex flow in detail. [05]
(b) Water flow over a rectangular weir 1.5 m at a depth of 10 cm and afterwards passes through a triangular right angled weir. Determine the depth of water through the triangle weir. Take discharge co-efficient for the rectangular and triangular weirs 0.63 and 0.58 respectively. [05]
(c) Derive an expression for discharge through a rectangular notch. [04]

SECTION – II

- Q - 4 (a) What is momentum? Explain moment correction factor. [02]
(b) State Bernoulli's theorem. [05]
The water is flowing through a tapering pipe having diameter 250 mm and 125 mm at section 1 and 2 respectively. The discharge through the pipe is 30 liters/s. The section 1 is 9m above datum and section 2 is 5m above datum. Determine the intensity of pressure at section 2 if that at section 1 is 450 kN/m^2 .
Q - 5 (a) Discuss Total energy line and Hydraulic gradient line with neat sketch. [05]
(b) Show by dimensional analysis that power P develops by a hydraulic turbine depends on mass density of liquid ρ , rotational speed N , diameter of runner D , working head H and the gravitational acceleration g . [05]
(c) What are various dimensionless numbers? Explain any two in detail. [04]

OR

- Q - 5 (a) Determine the head loss due to friction in a pipe of diameter 250 mm and length 60 m through which water is flowing at a velocity of 2.5 m/s using (i) Darcy's formula, (ii) Chezy's formula for which $C = 60$. Assume kinematic viscosity for water as 0.01 stoke. [05]
(b) Describe procedure to conduct dimensional analysis using Buckingham's Π -Theorem. [05]
(c) An oil of specific gravity 0.9 and viscosity 0.003 poise is required to transport through a pipe of 1 m diameter at the rate of $0.3 \text{ m}^3/\text{sec}$. Tests were conducted on a 10 cm diameter pipe using water at 20°C . The density and viscosity of water at 20°C are 1000 kg/m^3 and 0.001 poise respectively. Determine the average velocity and rate of flow in the model. [04]
Q - 6 (a) Define Mach Number, Mach cone and Compressible flow. [03]
(b) Derive Bernoulli's equation for compressible flow undergoing adiabatic process. [03]
(c) Describe in detail the displacement thickness, (δ^*) . [04]

- (d) Lubricating oil of specific gravity 0.85 and dynamic viscosity 0.1 Ns/m^2 is pumped [04]
through a 3 cm diameter pipe. If pressure drop per meter length of the pipe is 15 kPa,
determine (i) the mass flow rate, (ii) the shear stress at the pipe wall (iii) the Reynolds
number of flow.

OR

- Q - 6 (a) An aeroplane is flying at a height 15 km where temperature is -48°C . The Mach [02]
number is 2 corresponding speed of plane. Calculate the speed of the aeroplane.
Assume $R = 0.287 \text{ KJ/kg K}$ and adiabatic index $\gamma = 1.4$ for air.
- (b) State the equation of maximum velocity, average velocity and shear stress for laminar [03]
viscous flow between two stationary parallel plates.
- (c) An aeroplane is flying at 900 km/hr through still air having a pressure of 80 kN/m^2 [04]
and temperature of -8°C . Find the Mach number. Also find the stagnation properties
on the nose of the plane. Take $R = 287 \text{ J/kg-K}$ and $\gamma = 1.4$
- (d) Derive an expression for the velocity distribution for viscous flow through a circular [05]
pipe.
